Julio Cesar Aldana Almanza

Three-Tier Cloud Architecture Deployment

Agenda

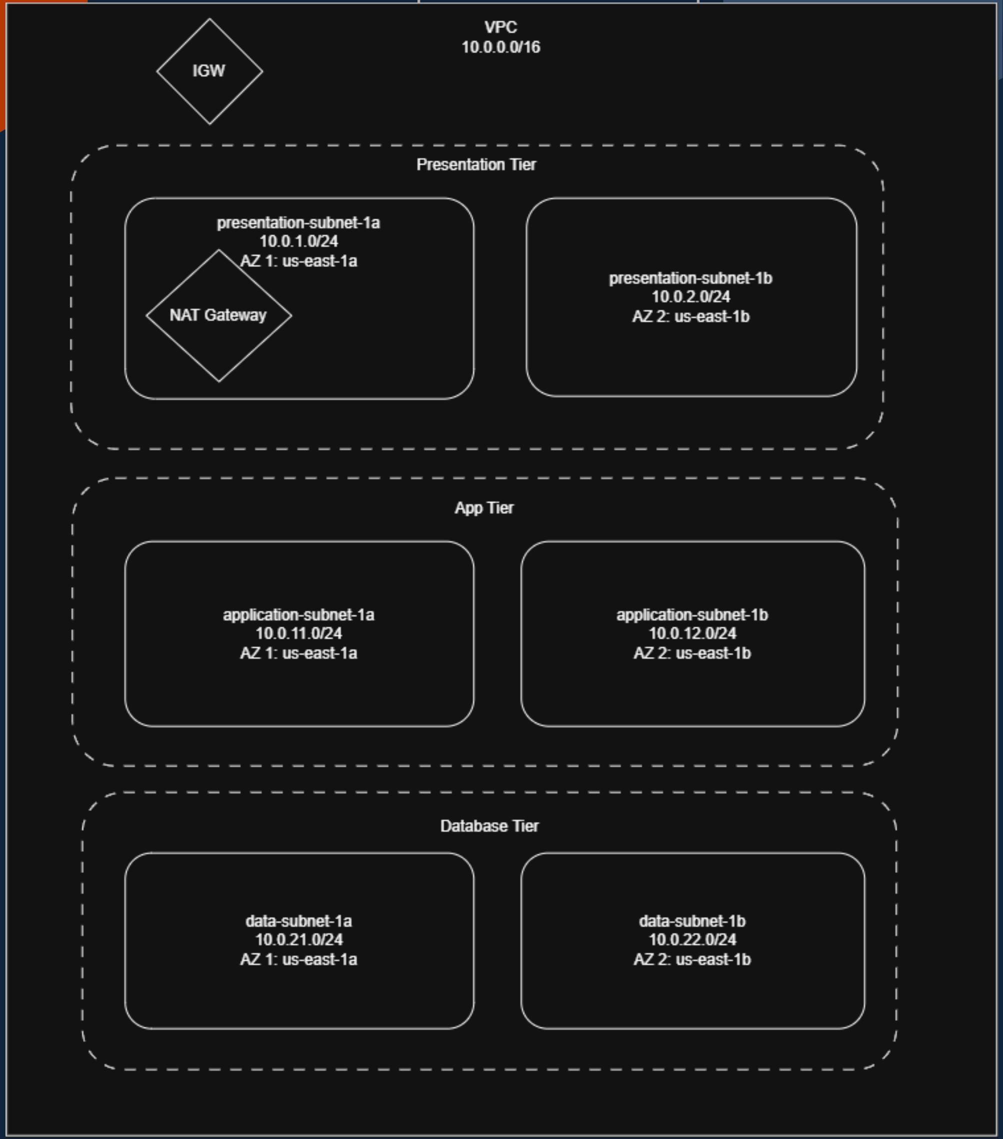
- Architecture Overview
- Challenges & Solutions
- Cost Analysis
- Improvements & Next Steps
- Live Demo

Three-Tier Architecture

Virtual Private Cloud

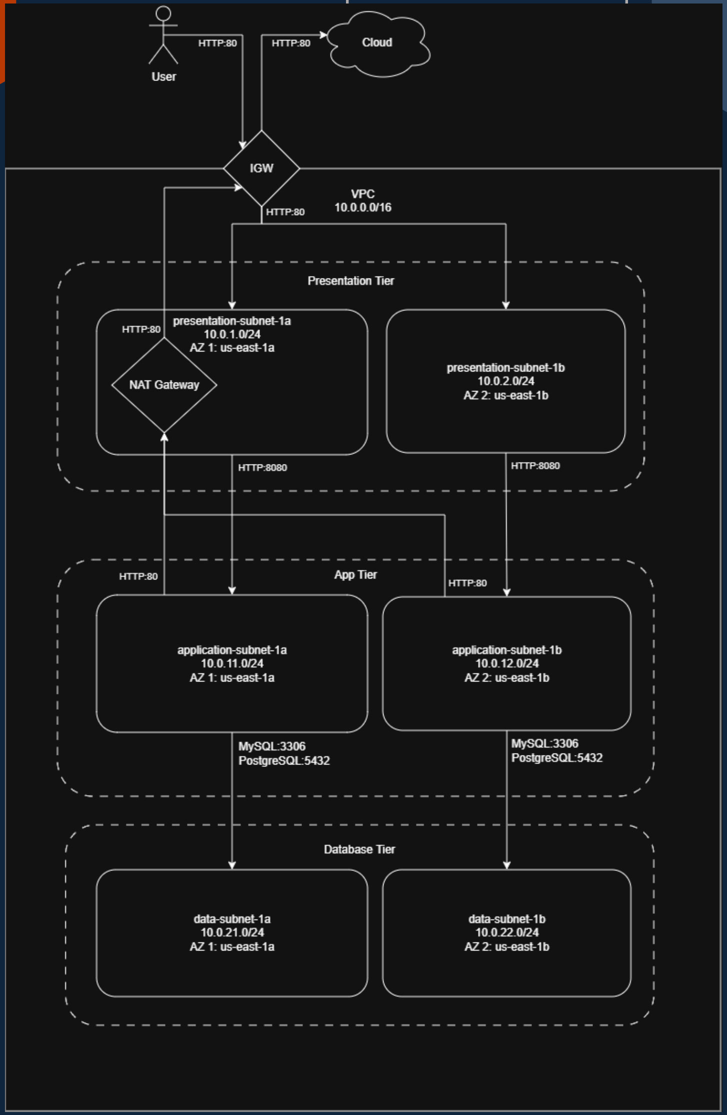
Network strategy

- VPC: 10.0.0.0/16
- Presentation Subnets (Public):
  - AZ 1: 10.0.1.0/24
  - AZ 2: 10.0.2.0/24
- Application Subnets (Private):
  - AZ 1: 10.0.11.0/24
  - AZ 2: 10.0.12.0/24
- Data Subnets (Private):
  - AZ 1: 10.0.21.0/24
  - AZ 2: 10.0.22.0/24



Traffic Flow

- Internet Gateway: allow traffic from outside to VPC
- Presentation Tier:
  - Allows HTTP requests from IGW (Port 80)
  - NAT Gateway attached to allow outbound requests from app-tier
- Application Tier:
  - Receives HTTP request from presentation tier only (Port 8080)
  - Outbound traffic allowed through NAT Gateway for package downloads
- Data Tier
  - Isolated from Outside: receives request from app tier only (MySQL:3306 and PostgreSQL:5432)



Security Groups

vpc-bastion

Direction	Port	Source / Destination	Purpose
Inbound	22	92.209.213.67/32	SSH from my laptop
Outbound	22	vpc-data-tier-sg, vpc-app-tier-sg	Hop to tiers for debugging purpose

vpc-alb-sg

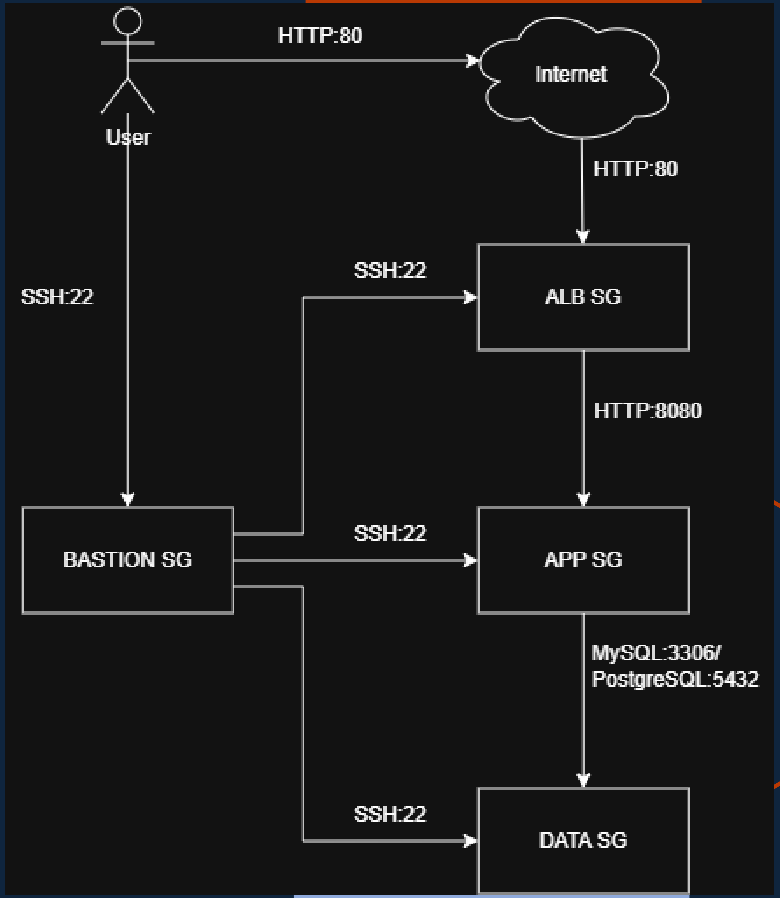
Direction	Port	Source / Destination	Purpose
Inbound	80	0.0.0.0/0	Public HTTP (internet-facing)
Inbound	443	0.0.0.0/0	Public HTTPS
Outbound	All	0.0.0.0/0	Package downloads

vpc-data-tier-sg

Direction	Port	Source / Destination	Purpose
Inbound	3306	sg-app-tier	MySQL from app tier only
Inbound	5432	sg-app-tier	PostgreSQL from app tier only
Inbound	22	sg-bastion	SSH via bastion only
Outbound	443	0.0.0.0/0	Package downloads

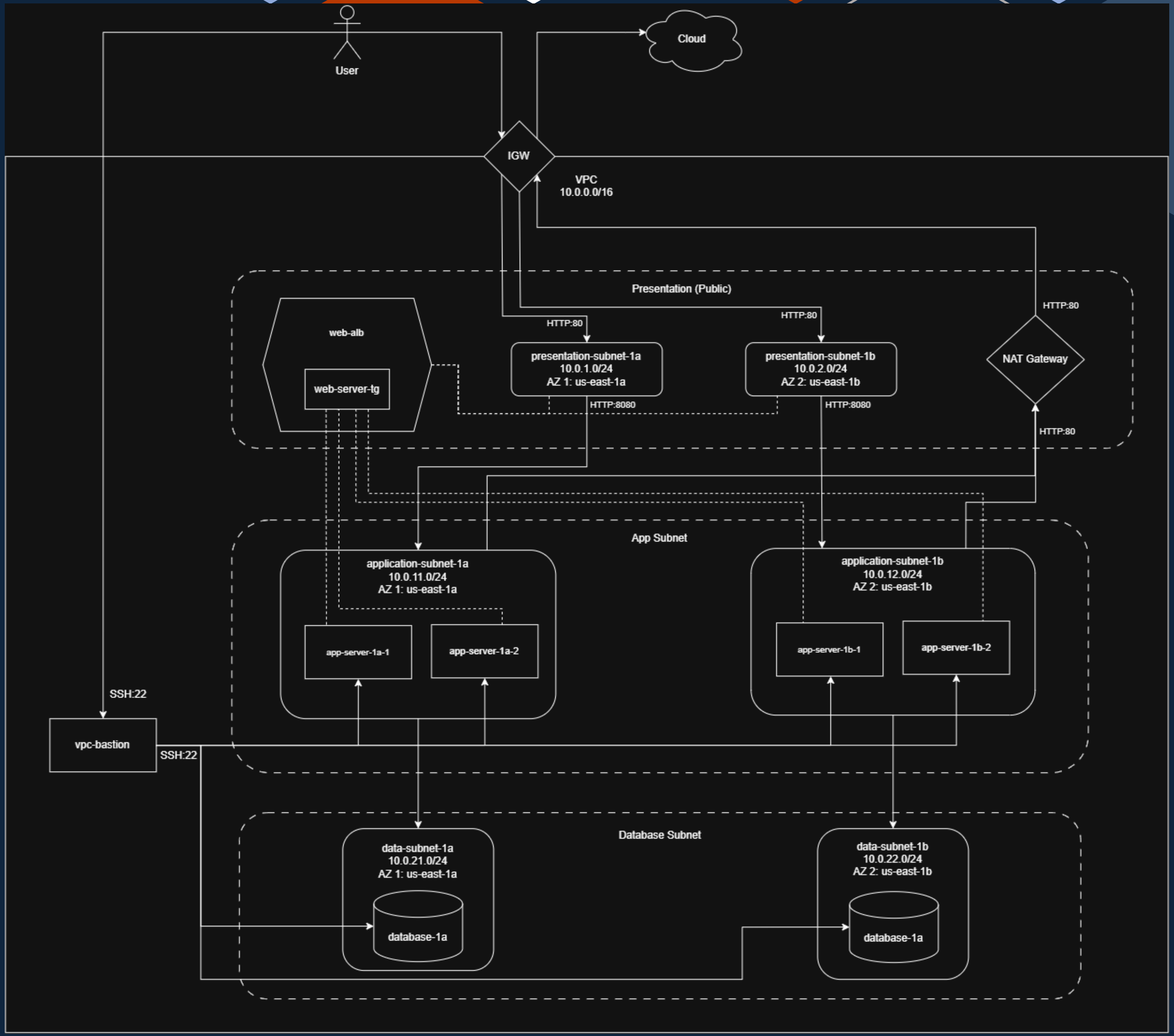
vpc-app-tier-sg

Direction	Port	Source / Destination	Purpose
Inbound	8080	vpc-alb-sg	App traffic from ALB
Inbound	80	vpc-alb-sg	App traffic from ALB
Inbound	22	vpc-bastion	SSH via bastion only
Outbound	3306	sg-database	MySQL queries
Outbound	5432	sg-database	PostgreSQL queries
Outbound	443	0.0.0.0/0	Package downloads



Three-Tier Architecture

- Presentation Subnets
- Application Subnets
- Data Subnets
- Bastion
- Internet Gateway
- NAT Gateway
- Load Balancer
- Target Group



Challenges & Solutions

Application Instance stars unhealthy

Application server is marked as unhealthy at the first start.

```
  throw err;
  ^

Error: Cannot find module 'pg'
Require stack:
- /home/ec2-user/server.js
    at Module._resolveFilename (node:internal/modules/cjs/loader:1140:15)
    at Module._load (node:internal/modules/cjs/loader:981:27)
    at Module.require (node:internal/modules/cjs/loader:1231:19)
    at require (node:internal/modules/helpers:177:18)
    at Object.<anonymous> (/home/ec2-user/server.js:3:20)
    at Module._compile (node:internal/modules/cjs/loader:1364:14)
    at Module._extensions..js (node:internal/modules/cjs/loader:1422:10)
    at Module.load (node:internal/modules/cjs/loader:1203:32)
    at Module._load (node:internal/modules/cjs/loader:1019:12)
    at Function.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:128:12) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [ '/home/ec2-user/server.js' ]
}

Node.js v18.20.8
```

Solution

Run bash `npm install pg` to install the dependency needed.

Run bash `sudo chmod 777 app.log` to allow edition of app.log file.

Run the application using the command `sudo nohup node server.js > app.log 2>&1 &`.

Application not running on restart

Application is not running again after restarting. A manual restart of the application is needed..

Solution

```
sudo npm install -g pm2   # install PM2 globally so the `pm2` command is available system-wide

mkdir app
mv server.js app/server.js

cat > ~/app/ecosystem.config.js <<'EOF'
module.exports = {
  apps: [{
    name: 'server-app',                              // the name PM2 shows in `pm2 list`
    script: 'server.js',                             // the file PM2 runs
    env: { NODE_ENV: 'production', PORT: 8080 }       // environment variables passed to the app
  }]
};
EOF

cd ~/app
pm2 start ecosystem.config.js   # start the app defined in the ecosystem file (applies the env block)
pm2 list                        # show all PM2-managed processes and their status

pm2 startup
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u ec2-user --hp /home/ec2-user
pm2 save                            # snapshot the current process list so it is restored on boot
systemctl is-enabled pm2-ec2-user   # confirm the PM2 boot service is registered; should print: enabled
```

Host key verification failed

Database connection through bastion refused due to warning "Remote host identification has changed!"

```
PS C:\Users\jcald\ce-project-three-tier-arch> ssh db
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!     @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:4HB3nrAiJ7B6cvpLhQu/JjKUwSwV9MatEmeoUsTgs5I.
Please contact your system administrator.
Add correct host key in C:\\Users\\jcald/.ssh/known_hosts to get rid of this message.
Offending ECDSA key in C:\\Users\\jcald/.ssh/known_hosts:43
Host key for 10.0.21.10 has changed and you have requested strict checking.
Host key verification failed.
```

Solution

Run `ssh-keygen -R DB-PRIVATE-IP` to remove any key that the db IP address may have attached.

Database simulation failed

Database instance is not running simulated responses

Solution

- NCat was not installed properly when running the script in the instance creation.
- If the instance is already created, run bash sudo dnf install nmap-ncat to install the dependency.
- To solve the issue permanently (create a healthy database instance), modify the userdata script as follows:

```
#!/bin/bash
# This simulates a database server
# In production, you'd use RDS, not EC2

sudo yum update -y
sudo yum install -y nc netcat
sudo dnf install -y nmap-ncat
```

Cost Analysis

Monthly Cost Breakdown

Service	Quantity	Calculation	Monthly Cost (USD)
VPC	1	1 × $0/month	$0
Internet Gateway	1	1 × $0/month	$0
Subnets	6	6 × $0/month	$0
Route Tables	4	4 × $0/month	$0
EC2 t3.micro instances	4	4 × $0.0104/hr × 730 hrs	$30.37
EC2 EBS gp3 storage	4	4 × 8 GB × $0.08/GB-month	$2.56
Application Load Balancer	1	1 × $0.0225/hr × 730 hrs	$16.43
NAT Gateway hourly cost	1	1 × $0.045/hr × 730 hrs	$32.85
NAT Gateway data processing	100 GB	100 GB × $0.045/GB	$4.50
RDS db.t4g.micro compute	2 ins.	2 × $0.017/hr × 730 hrs	$24.82
RDS gp3 storage	100 GB	100 GB × $0.115/GB-month	$11.50
RDS backups	NA	Included within allocated storage	$0
CloudWatch basic metrics	NA	Included	$0
VPC Flow Logs	NA	NA	~$5

Total monthly cost : ~$128 USD

Cost optimization strategies

1. Enable auto-scaling to launch only sufficient instances to keep the system working.
2. Add VPC Endpoints for services such as RDS, Systems Manager, CloudWatch to reduce NAT Gateway usage.
3. Apply Saving Plans for stable EC2 workloads to reduce On-demand costs.
4. If acceptable for non-critical workloads, run a Single-AZ RDS instance instead of Multi-AZ (roughly halves the RDS compute cost)
5. Optimize logging retention
6. Revisit instance sizes after collecting 30–60 days of CloudWatch metrics

Improvements & Next Steps

Improvements

Short-term

1. Replace the bastion host with AWS Systems Manager Session Manager

   Benefits: No inbound ports, no public IP, IAM-based access, smaller attack surface, no SSH key management

2. Replace database placeholder with RDS Database

3. Update health checks in application to monitor CPU Utilization

4. Add Logging: CloudTrail, ALB Access Logs, NAT Gateway Metrics to improve troubleshooting

5. Remove unused rules in Security Groups

Long-term

1. Add VPC Endpoints: Create Interface/Gateway Endpoints for services like S3, DynamoDB, Systems Manager, Secrets Manager

   Benefits: Less NAT Gateway Traffic, Lower cost, private AWS Service access

2. Add Auto-scaling por application tier

3. Increase security with Zero Trust/Least privilege principle: IAM authentication, Session Manager, service-to-service authentication.

4. For stricter security requirements, use services like AWS Network Firewall, Stateful Inspection, Domain filtering

Live Demo

Q&A

Thank you

Julio Cesar Aldana Almanza